Holant lead, step (b): digit-level (QTT) structure of the decomposition transfer matrix

Lead: LEAD-EXO-holant-decomposition-count-network, next_action step (b). Protocol: protocol.json, HOLANT-STEPB-QTT-20261104, frozen before data (sha256 bdba447e…4cd3, commit 4970366b). The run and analysis scripts are frozen by hash in it.

Verdict (frozen rule, primary metric): NULL, controls valid. At $p \le 2^{10}$ the transfer matrix M has no digit-level tensor-train structure in its natural x-coordinate labelling beyond what a random relabelling of F_p has. This is a structure statistic, not an attack-cost claim, and no comparison against rho is made.

Why step (b) was reformulated

The row asked for the tensor rank, or flattening ranks, of T after bond-wise basis changes (additive Fourier, multiplicative characters, Legendre bases), compared with a random-support null. Every flattening rank, and the CP rank, is invariant under invertible bond-wise maps: the mode-i flattening of $(A \otimes B \otimes C)\cdot T$ equals $A \cdot T_{(i)} \cdot (B \otimes C)^T$, whose rank is rank $T_{(i)}$ when A, B and C are invertible. Holographic transformations therefore cannot lower these ranks. invariance.py confirms this at $p = 61$: all three flattening ranks are 61, in the original basis and after Fourier, character and random invertible changes. Only the non-invertible Legendre projection drops a rank, to 2, which is trivial.

The question that matters for an algorithm polynomial in log p is whether the F_p alphabet, split into binary digits, gives M a low **quantized tensor-train (QTT)** rank. That measure depends on the labelling, so the test compares labellings.

Design (frozen)

- **Object.** $M[y,z] = \#\{x \in FB : S_3(x,y,z) = 0\}$ over the full alphabet F_p, zero-padded to $2^L \times 2^L$. FB is the $\lfloor\sqrt{p}\rfloor$ smallest x-coordinates of $E(F_p)$. Sizes $L = 6..10$, with p the largest prime below 2^L. There are 3 curves per size, drawn from a protocol-hash-seeded stream, each with #E an odd prime.
- **Primary metric.** The largest numeric rank over the $2L-1$ unfoldings with interleaved digits (y_msb, z_msb, …, y_0, z_0).
- **Arms.**
  - X: the natural x-labelling (the candidate).
  - N: 99 uniform random relabellings per curve (the null). Its two seed streams by parity form the A/A check.
  - LOG: the E-block relabelled by discrete-log orbit index (positive control).
  - SH: the shift matrix $[z = y + 1]$ (detector check).
- **Rule.** SIGNAL requires X below every null draw on all 9 curves with $L \ge 8$, and $X \le 0.75 \times$ the null median at $L = 10$. NULL requires X inside the null's 5th–95th percentile on at least 7 of those 9 curves. Validity requires LOG below the null minimum on all 9 curves, and the A/A median difference to be at most the pooled IQR.

Results

All 1530 matrices were computed (exit 0, run.log). Applying the frozen analyze.py gives:

L	p	#E (3 curves)	X r_max	null min–max	LOG r_max	SH
6	61	61, 67, 67	48, 60, 64	46–64	25, 25, 25	5
7	127	113, 139, 107	120, 127, 127	110–128	25, 36, 31	5
8	251	263, 257, 271	256, 256, 256	252–256	49, 44, 57	5
9	509	479, 487, 523	512, 512, 512	509–512	76, 77, 78	5
10	1021	1031, 1049, 1049	1024, 1024, 1024	1023–1024	121, 109, 112	5

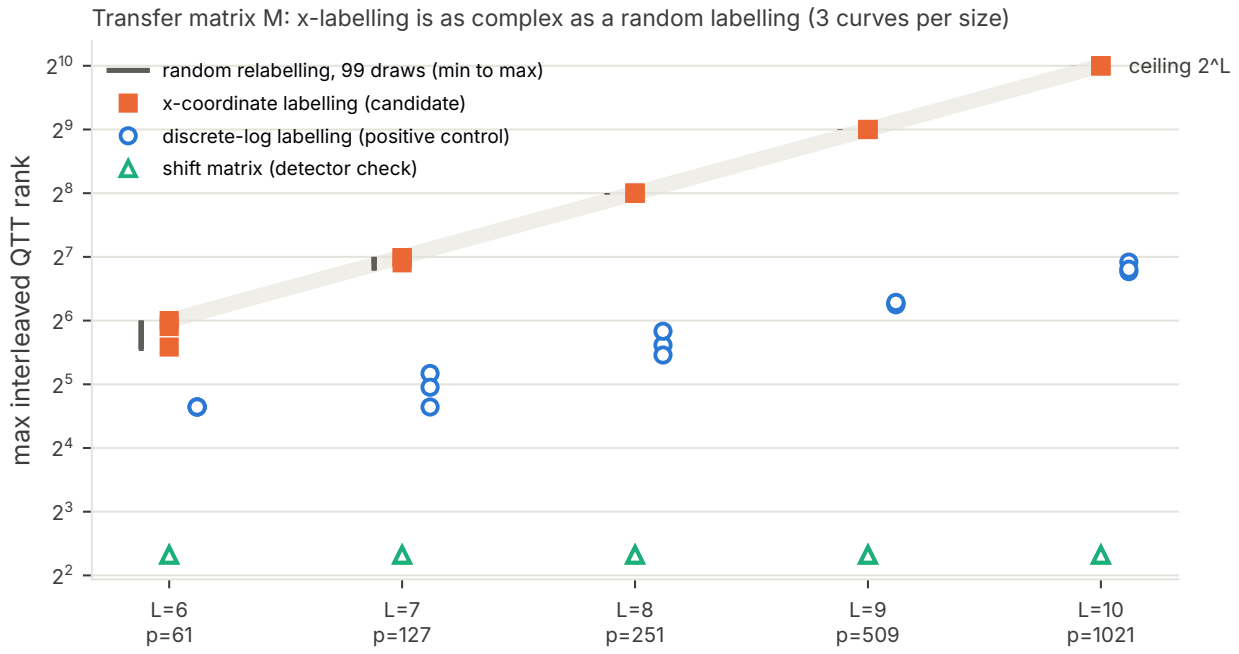

Figure 1: Interleaved QTT rank by arm

- **Verdict: NULL.** 9 of 9 curves with L ≥ 8 lie inside the null's central 90%. Both validity checks pass: LOG is far below the null minimum on every curve, and the A/A median differences are 0. From L = 8 onwards, X and the null both sit at the rank ceiling 2^L. The secondary parameter count Σ log2 rank also equals its maximum L^2 (64, 81, 100 bits) for X and every null draw.
- The positive control shows the detector works: the discrete-log labelling has rank 2.6–9.4 times below the ceiling (64/25 at p = 61 up to 1024/109 at p = 1021).
- At L = 6 one curve has X = 48 against a null median of 59 (one-sided p = 0.05). Sizes below 8 are outside the decision rule by design. At larger sizes the effect is absent.

Post-data observations (not preregistered)

1. **The secondary "blocked" metric is uninformative by construction.** Its maximum is at the middle cut, which is the ordinary matrix rank of M (for example 524 at p = 1021). A permutation cannot change that rank, so X, LOG and every null draw are identical there. This is why verdict_blocked.json reports INVALID_CONTROLS. It is a defect in the protocol's metric choice, not a finding about M.
2. **Even the ideal labelling has rank of order √p.** LOG r_max grows from 25 to about 115 between p = 61 and p = 1021, roughly 3.6√p. In LOG order M is Toeplitz-plus-Hankel with about 2|FB| ≈ 2√p diagonals whose positions are effectively random, so its QTT rank grows with |FB|. If some labelling were as good as knowing the logarithms, a QTT contraction would still carry rank of order √p, the same scale as rho. This bounds what any labelling-based route through this network can gain.

Amendment A1 (filed 2026-11-04, after the data; does not affect the verdict)

tools/experiment_protocol.py check-export reports 1547 findings on export.json.gz. They come from two encoding errors in the frozen protocol, found only when exporting:

- sample_size says 15 instances × 1 replicate, which does not describe the 99 null draws per curve.
- The run script derives seeds as sha256(protocol_sha | tag | L | curve | draw), as the protocol's seed_derivation.note states. The checker's seed table uses its own formula.

The run executed exactly the frozen script (hash-checked), and every seed is reproducible. protocol.json is left unchanged, because amendments are inside the freeze hash. A follow-up protocol should encode null draws as replicates and use the checker's derive_seed.

Deviations

- The resume checkpoint was not pushed to S3, because the container has no upload token. With the owner's approval, the pushed branch served as the resume record.
- rows.jsonl (569 KB) and export.json (1.6 MB) are kept as gzip files. The session cannot write to the artifact store, and the compressed files are under the 64 KiB threshold. Uncompressed hashes are in data.sha256.

Consequence for the lead

Step (b) as originally worded is ill-posed, by the invariance above. In its reformulated digit-QTT form it gives NULL at $p \le 2^{10}$, and even an oracle labelling caps the rank at order $\sqrt{p}$. Per the row's discriminator, the lead becomes exhausted_at_scale for the QTT measure at $p \le 2^{10}$. It reopens only for a basis or labelling family whose description grows sub-polynomially in p and gives rank below $\sqrt{p}$. The positive control shows that even the discrete-log labelling does not.

Reproduce

```
zcat rows.jsonl.gz > rows.jsonl      # or rerun:
OMP_NUM_THREADS=1 python3 qtt_rank.py --protocol-sha
bdba447e899094f553eb43b857230191e0b5f8d494988d9cf9c4d6b577b34cd3 --out rows.jsonl
python3 analyze.py rows.jsonl verdict.json interleaved
python3 analyze.py rows.jsonl verdict_blocked.json blocked
python3 figures.py; python3 invariance.py; ./build_pdf.sh
```

The run took about 18 minutes on one core of the session container. The ranks are exact counts, and no clock is measured.